

RAM, ROM & Cache

Lesson 08 — Volatile and Non-Volatile Memory

JKSSB Computer Awareness

October 8, 2026

Two Kinds of Memory

- ▶ Main memory holds the data and instructions the CPU uses directly.
- ▶ The key split is **volatile** versus **non-volatile**.
- ▶ **Volatile**: contents are lost when power is removed.
- ▶ **Non-volatile**: contents are retained without power.

Trap (TRAP-01)

RAM is volatile; ROM is non-volatile. Do not reverse the two.

RAM — Volatile Main Memory

- ▶ **RAM** is **Random Access Memory**.
- ▶ It is the **main memory** that holds running programs and data.
- ▶ **Volatile**: on power-off, its contents disappear.
- ▶ “Random access” means any location is reached in roughly equal time.
- ▶ In the office desktop (EX-01), the OS, MS Word, and the spreadsheet load into RAM while in use.

Exam cue

RAM = working space; cleared when the machine is switched off.

Two Types of RAM: DRAM and SRAM

DRAM

- ▶ Dynamic RAM
- ▶ Needs periodic **refresh**
- ▶ Denser and cheaper
- ▶ Used for **main memory**

SRAM

- ▶ Static RAM
- ▶ No refresh needed
- ▶ **Faster**, but costlier
- ▶ Used for **cache**

Do not confuse

Both are RAM and both are volatile — they differ in speed, cost, and role.

DRAM vs SRAM at a Glance

Feature	DRAM	SRAM
Full form	Dynamic RAM	Static RAM
Refresh	Needs periodic refresh	No refresh
Speed	Slower	Faster
Density / cost	Denser, cheaper	Less dense, costlier
Typical use	Main memory	Cache memory

DRAM: many cheap cells. SRAM: fewer, faster cells.

ROM — Non-Volatile and Read-Only

- ▶ **ROM** is **Read-Only Memory**.
- ▶ **Non-volatile**: it keeps its contents without power.
- ▶ **Read-only in normal operation**; writing needs a special programming process.
- ▶ It commonly stores **firmware** such as the start-up code.
- ▶ RAM and ROM are both primary memory; they differ in volatility and writeability.

Trap (TRAP-02)

ROM is not freely read and write. Normally you can only read it.

The ROM Family

Type	Key property
PROM	Programmable once; cannot be erased
EPROM	Erased by ultraviolet (UV) light
EEPROM	Erased and rewritten electrically
Flash	EEPROM-derived; erased in blocks

PROM once EPROM by UV EEPROM electrically Flash in blocks.

Cache — Fast Memory Near the CPU

- ▶ **Cache** is a small, very fast memory between the CPU and main memory.
- ▶ It holds frequently or recently used data and instructions.
- ▶ Purpose: reduce the CPU's wait for slower main memory.
- ▶ Cache is **faster than main memory** (PYQ-03).
- ▶ Cache is working memory, **not** a backup or permanent store.

Exam cue

Speed order: cache > main memory > secondary storage.

Cache Levels: L1, L2, L3

- ▶ **L1**: smallest and fastest, closest to the core.
- ▶ **L2**: larger and a little slower than L1.
- ▶ **L3**: largest and slowest of the three, typically shared across cores.
- ▶ All cache levels are faster than main memory.

Exam cue

Size and distance: **L1 < L2 < L3**. Speed: reverse.

Cache Hit, Miss, and Hit Ratio

- ▶ **Hit**: the CPU finds the required data in the cache.
- ▶ **Miss**: the data is not in the cache, so it is fetched from main memory.
- ▶ **Hit ratio** = hits $\div$ total accesses.
- ▶ Example: 100 accesses with 90 hits give a hit ratio of $90/100 = 0.90$.
- ▶ A higher hit ratio means less time waiting for main memory.

RAM vs ROM vs Cache

Aspect	RAM	ROM
Volatility	Volatile (TRAP-01)	Non-volatile
Normal access	Read and write	Read-only (TRAP-02)
Typical role	Main memory	Firmware / start-up code
Speed	Faster than secondary	Slower than cache

Cache sits between the CPU and RAM and is the fastest of the three.

Common Traps in This Topic

- ▶ **TRAP-01**: RAM is volatile; ROM is non-volatile.
- ▶ **TRAP-02**: ROM is read-only in normal operation.
- ▶ Cache is **faster** than main memory, not slower (PYQ-03).
- ▶ **SRAM** goes to cache; **DRAM** goes to main memory.
- ▶ Cache is **not** a backup — it is temporary working memory.

Lesson 08: Recall

- ▶ RAM is volatile; ROM is non-volatile and normally read-only.
- ▶ DRAM: denser/cheaper, needs refresh, main memory; SRAM: faster, no refresh, cache.
- ▶ ROM family: PROM (once), EPROM (UV), EEPROM (electrical), flash (block erase).
- ▶ L1 smallest/fastest, L3 largest/shared; all faster than RAM.
- ▶ Hit ratio = hits $\div$ total accesses.